Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Atlantic Meridional Overturning Circulation

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QUID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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I INTRODUCTION

I.1 Product name

GLOBAL_OMI_AMOC_26n_profile, GLOBAL_OMI_AMOC_max26n_timeseries

I.2 Indicator evaluation method

The Atlantic Meridional Overturning Circulation (AMOC) is calculated at 26.5°N by integrating the meridional velocity *v* across the Atlantic along the nearest latitude line to 26.5°N and then doing a cumulative integral in depth. Hence the AMOC profile *P* each month is given by:

$$P(z,t) = \int_0^z \int_{east}^{west} v(x,z',t)\, dx\, dz'$$

From this the monthly AMOC time series *M* is calculated as the maximum of *P* in depth each month. A climatological profile is calculated by taking the time mean of *P* from 1993 to the end of 2023.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
GLOBAL_MULTIYEAR_PHY_ENS_001_031	QUID : Foreword (copernicus.eu) PUM: CMEMS-GLO-PUM-001-031.pdf (copernicus.eu)

A comparison is made to an observational product (RAPID) of the AMOC which started in April 2004. Details of the observational product can be found in *Moat B.I. et al, 2022* and the data downloaded from Smeed et al (2017) https://rapid.ac.uk/rapidmoc/rapid_data/datadl.php.

II SCIENTIFIC VALIDATION

A multi-product approach is used to construct OMIs for the AMOC (see Jackson et al, 2018). Four ocean reanalyses (GloSea5, ORAS5, C-Glorsv7, Glorys2v4) are used to calculate time series of the AMOC (*M*) and climatological mean profiles (*P*). These four reanalyses are then combined to form the Global Reanalysis Ensemble Product (GREP) with an ensemble mean and spread (2 x ensemble standard deviation) and compared to an observational product (RAPID).

Results in Fig 1 show a profile that is mostly consistent with the observational product over the same time period. All individual products and the GREP show too little volume transport near the surface resulting in a profile that is weaker that than the observations in the upper 800m. The maximum of the profile agrees well with the observations and the deep flow is consistent with the observations, although this is a result of the agreement of one reanalysis (CGLORS) while the others have a too shallow return flow. It should be noted, however, that significant differences in the deep profile from RAPID observations are expected because of the observational methodology (Roberts et al, 2013; Stepanov et al 2016).

The timeseries of *M* (Fig 2 top panel) also show agreement with the observations, with a correlation of 0.82, root mean square error of 2.0 Sv (1 Sv = 10^6 m^3/s) and bias of 0.43 Sv for the monthly values of the multi-product over the observational period (Apr 2004-Mar2016). Applying a 12 month running mean removes the seasonal cycle and high frequency variability (Fig 2 bottom panel). Again this shows good agreement with the RAPID observations over their common period with a root mean square error of 0.8 Sv for the multi-product and annual mean values have a correlation of 0.96. The multi-product captures important features of the observations with a temporary decrease in winter 2009-2010 (McCarthy et al, 2012) and a decadal weakening of the circulation (Smeed et al, 2014; Jackson et al, 2016), although the magnitudes of change are smaller than in the observations. There is more spread amongst the reanalyses earlier on, particularly from 1993-1998 where one reanalysis (ORAS5) has a strong AMOC initially followed by a large weakening. Given that this signal is not seen in the other reanalyses this leads to a large spread in the GREP.

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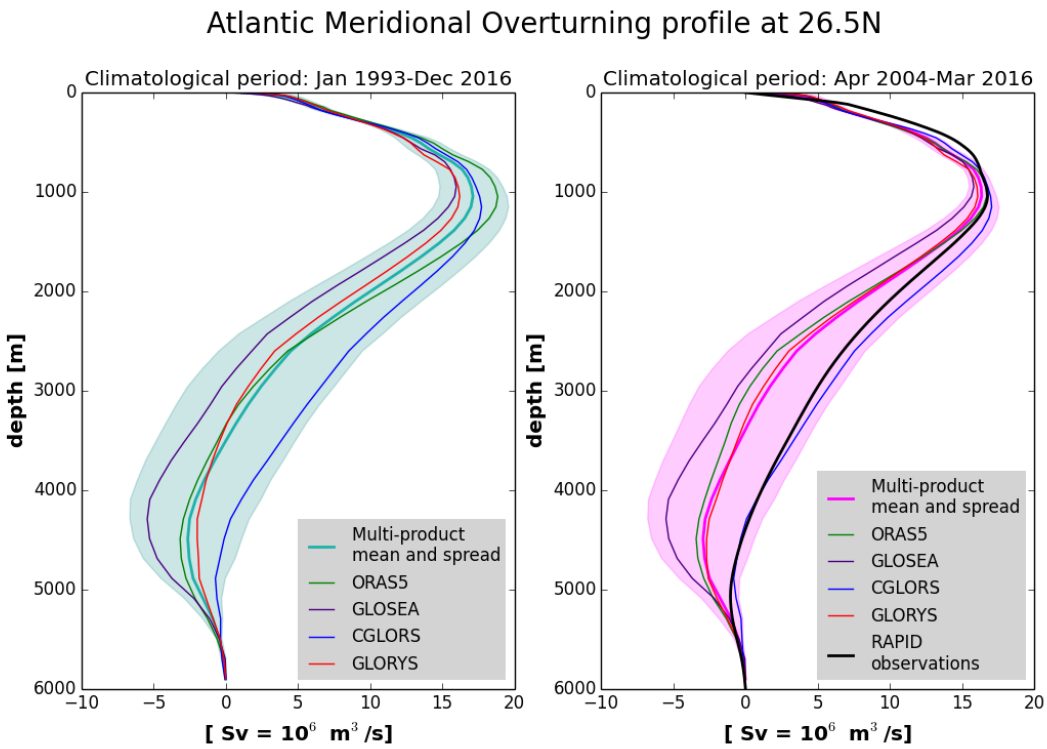

Figure 1: AMOC climatological mean profiles *P* for two time periods. Lines show the individual reanalysis products with the multi-product mean shown with the thick cyan (left) or magenta (right) line and 2 x the multiproduct standard deviation shown with shading. The right panel also shows observations from the RAPID array (Smeed et al, 2017).

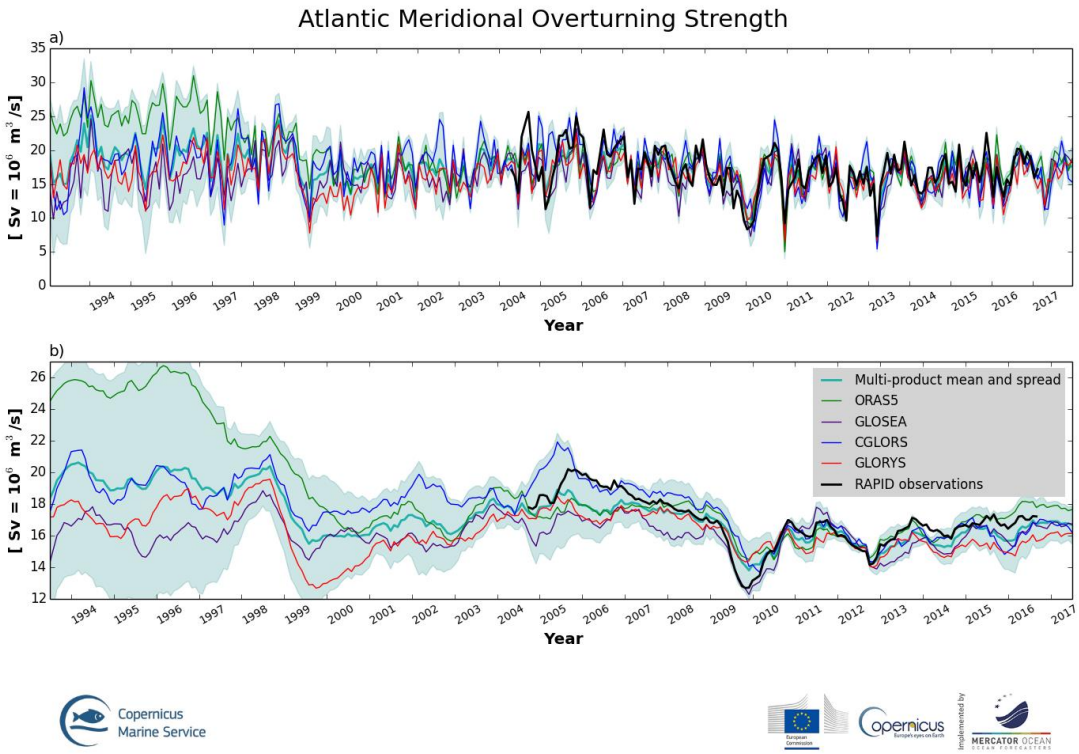

Figure 2 : AMOC timeseries M. Lines show the individual reanalysis products with the multi-product mean shown with the thick cyan and 2 x the multiproduct standard deviation shown with shading. The black line shows M from the RAPID array (Smeed et al, 2017).
Top: monthly mean values. Bottom: Monthly mean values with a 12-month running mean applied.

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